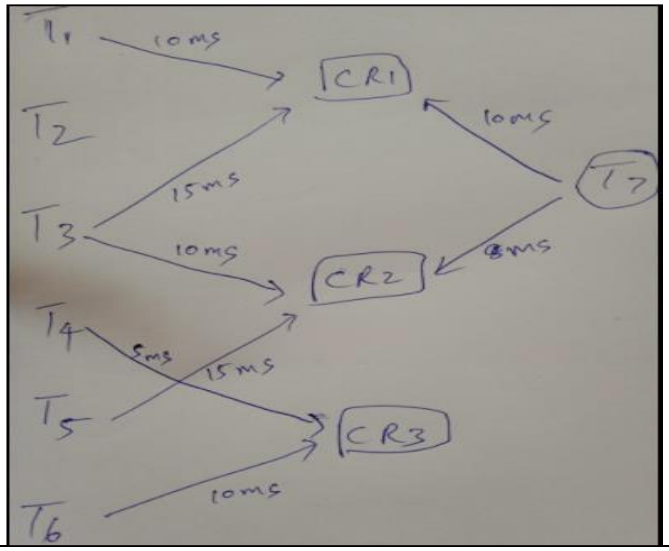
MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

SEVENTH SEMESTER B. TECH (ELECTRONICS AND INSTRUMENTATION)
PROCTORED ONLINE MAKE UP EXAMINATION Feb 22

SUBJECT: **REAL TIME OPERATING SYSTEM (ICE 4060)**

TIME: 2.20 PM to 3.45 PM DATE: 17-2-2022 MAX MARKS 20

Note: Submit your responses (answer script PDF file) before scheduled time.

1.	a	With an example and timing diagram explain the following terms a. Relative and absolute dead line b. Response time	3
	b	Explain different types of priority Inversion available under PCP with an FSM diagram. Also draw the Priority inversion table for the following problem 	4
	c	Explain the procedure of synchronization done in presence of byzantine clock. Show how clock time differs when drived with 3 clock set. Also compute resynchronization period for the same	3
2.	a	Consider the following three periodic real time task T1- (10,22), T2 – (12,55) and T3 (18,100). To run in a uniprocessor, determine whether the task is schedulable under, RMA or DMA.	4
	b	Explain the SR modelling timing constraint for a washing machine. Also draw its FSM	3
	c	With an example and timing diagram explain the following terms a. Task precedence b. Data sharing	3